

Software Management: Experiment #1

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Problem 1

Market Simulation Experiment Report

Experimental Background

This experiment simulated a perfectly competitive market for a backpack. The primary objective was to derive demand and supply curves through a series of auctions, providing insights into market behavior. The instructor served as the supplier, while students participated as market participants submitting bids based on their willingness to pay.

The experiment aimed to demonstrate fundamental principles of microeconomics, including:

- How individual preferences aggregate to form market demand
- The relationship between price, quantity, and revenue
- The concept of profit maximization
- The significance of the equilibrium price point

Experimental Content and Steps

The experiment followed a structured process:

1. The instructor presented the rules of the auction, specifying that the item was a backpack with no actual transaction costs.
2. As a participant, I evaluated my personal willingness to pay for the backpack, considering factors such as utility, personal budget constraints, and perceived value of the item.
3. All students submitted sealed bid cards within the specified price range (0-500 CNY), including student ID and name.
4. The instructor collected and tallied all valid bids (those between 0-500 CNY), ensuring basic anonymization of the data.
5. The bid data was organized in descending order by price to generate a demand table.
6. Using this demand table, demand and revenue curves were plotted to identify the optimal price point for profit maximization.
7. The winning bidder(s) corresponding to the optimal price were announced.

Experimental Results and Analysis

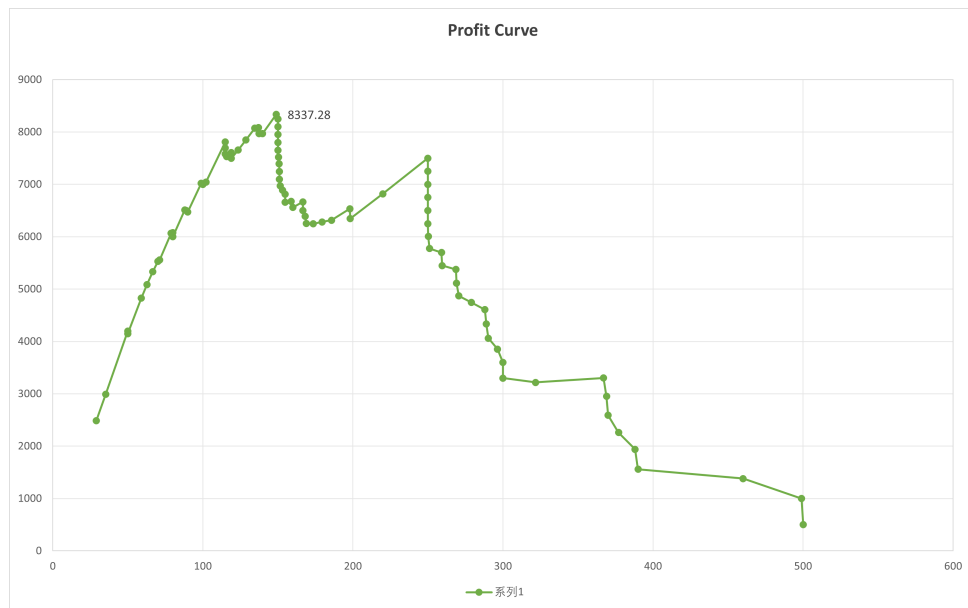


Image 1: Profit Curve

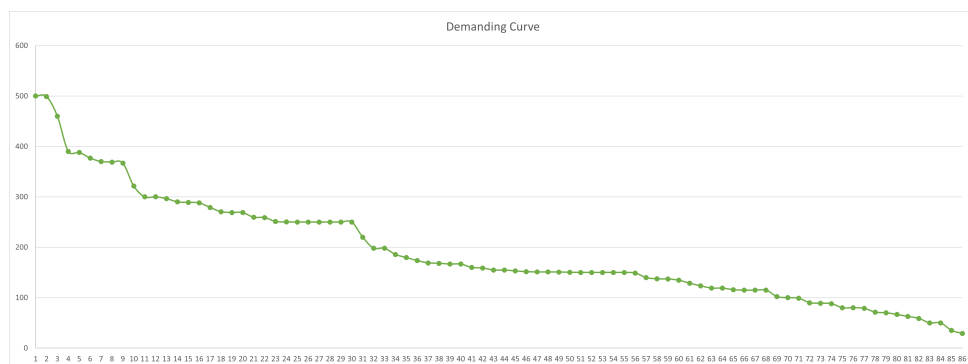


Image 2: Demanding Curve

Demand Curve Analysis

The demand curve (second graph labeled "Demanding Curve") displays a typical downward-sloping pattern, confirming the economic principle that quantity demanded increases as price decreases. Key observations:

- The curve begins at approximately 500 CNY with very low quantity demanded
- The curve exhibits several distinct segments with different elasticities:
 - A steep initial decline (elastic demand) between points 1-8
 - A more gradual decline between points 9-30
 - Several small plateaus where quantity remains constant across small price changes
 - A final gradual decline toward the maximum quantity
- The demand is highly elastic at higher prices and becomes increasingly inelastic at lower prices

Profit/Revenue Curve Analysis

The first graph labeled "Profit Curve" shows the total revenue at each price point. The revenue is calculated as $\text{price} \times \text{quantity demanded}$ at that price. Notable features include:

- The curve reaches its peak at 8337.28 (units not specified, presumably CNY) around the 130th bid
- There are several local maxima in the curve, including peaks around positions 80, 130, and 250
- After reaching the global maximum around position 130, revenue generally decreases with some fluctuations
- The sharp drops in the revenue curve correspond to significant price decreases in the demand curve

My Bid

My personal bid for the backpack was 310 CNY. I determined this value based on:

- The estimated market value of a standard backpack
- My personal utility for a new backpack
- My budget constraints as a student
- The knowledge that this was a simulated market

Based on the profit curve, the optimal price point appears to be around 130-140 CNY, which would maximize revenue at approximately 8337.28 CNY. At this price, approximately 60-65 units would be sold.

Learning Experience

This experiment provided several valuable insights:

1. **Practical Understanding of Economic Theory:** The experiment transformed abstract concepts like demand curves and profit maximization into tangible, observable phenomena. Seeing how individual preferences aggregated into a market demand curve was particularly enlightening.
2. **Strategic Bidding:** I learned that determining my bid required balancing my true valuation against strategic considerations. This highlighted the concept of consumer surplus - the difference between what I was willing to pay and what I actually might pay in equilibrium.
3. **Market Efficiency:** The auction mechanism efficiently processed information from multiple participants to determine a market-clearing price. This demonstrated how markets can coordinate diverse preferences without central planning.
4. **Elasticity in Practice:** The varying slopes in the demand curve showed how price elasticity changes across different price ranges. The steeper sections indicated higher elasticity (more sensitivity to price changes), while flatter sections showed more inelastic demand.
5. **Revenue Optimization:** The existence of a clear peak in the profit curve illustrated why sellers must carefully consider pricing strategies rather than simply setting the highest possible price.

6. Data Visualization: The graphical representation of the demand and revenue data made abstract economic concepts more concrete and understandable.

This experiment reinforced the fundamental principle that markets tend toward equilibrium prices that balance the interests of buyers and sellers. The optimal price point on the profit curve represents the price where marginal revenue equals marginal cost (in this case, zero marginal cost since we're dealing with a simulated auction with no production costs).

In conclusion, this market simulation provided an engaging hands-on demonstration of microeconomic principles. It transformed theoretical concepts into practical experiences, deepening my understanding of market mechanisms and pricing dynamics.